# OSINACHI CHUKWUJAMA

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## PROFILE

Photonics master's student with a First Class degree in Mechanical Engineering, drawn to inverse problems in imaging, where physical forward models meet learned priors. I have derived and implemented diffusion and flow-matching models in PyTorch, deployed vision models on embedded GPUs, worked hands-on with spatial light modulators, and built optics simulations. I am looking for a summer research project on inverse problems, from optimisation methods to physics-based and generative reconstruction.

## RESEARCH INTERESTS

Inverse problems in imaging, diffusion and flow-based priors for reconstruction, physics-based and differentiable imaging models, optimisation for machine learning.

## EDUCATION

**Erasmus Mundus Joint Master, Intelligent Photonics (iPSRS)** September 2026 – 2028
Université Jean Monnet (France), University of Eastern Finland, Université Paris-Est Créteil
*Relevant coursework:* Fourier optics, digital image processing, interferometry, probability and signal analysis, scientific computing with Python.

**Bachelor of Engineering, Mechanical Engineering** November 2017 – August 2023
Federal University of Technology, Owerri (FUTO), Nigeria. CGPA: 4.56/5.00 (First Class),
96th percentile in the department; best graduating student in Industrial and Production Engineering.

## RESEARCH AND LABORATORY EXPERIENCE

**Generative Modelling from First Principles** 2026
*Self-directed research*

- Derived and implemented DDPM (forward/reverse process, ELBO, reparameterisation) and OT-conditional flow matching in PyTorch, covering training and sampling end to end.
- Studied the links between score matching (denoising and implicit variants, NCSN), flow matching and normalising flows; wrote a forward-diffusion noising tool in Go for synthetic data generation.
- Proposed a comparison of OT-CFM and DDPM on photonics/laser data (TPU Research Cloud application).

**Spatial Light Modulator (SLM) Laboratory Practical** 2026
*iPSRS M1 coursework* *Université Jean Monnet*

- Generated diffraction gratings and Fresnel lens phase patterns in MATLAB and displayed them on an SLM.
- Observed and analysed the resulting diffraction patterns, and computed the gratings' spatial frequencies from measurements.

**Surveillance Drone with Real-Time Object Detection** August 2023 – March 2024
*Undergraduate final project and follow-up, supervised by Prof. O.O. Obiukwu* *MEE, FUTO*

- Collected and annotated a three-class dataset, fine-tuned YOLOv5 (70% mAP) and ran live inference on a drone's camera stream via ONNX.
- Deployed the model on an NVIDIA Jetson Nano with TensorRT, cutting processing time by 44%.

**Optics Playground** In progress
*Personal project, pre-zeemax.vercel.app*

- Interactive 2D ray-tracing simulator with finite-size lenses, being extended into a Fourier optics bench.

## SKILLS

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|---|---|
| **Machine learning** | PyTorch: DDPM, flow matching, score matching, CNN/ResNet training, transfer learning; ONNX, TensorRT |
| **Mathematics** | Probability, variational inference (ELBO), optimal transport for flow matching, Fourier analysis |
| **Programming** | Python (proficient), MATLAB, Go, TypeScript/JavaScript; Git, Linux, CI/CD, automated testing |
| **Languages** | English: C1 (IELTS 8.0) |

## WORK EXPERIENCE

**Dstrct Group** March 2025 – May 2026
*Backend Engineer / Technical Lead* *Lagos, Nigeria*

· Technical lead on an enterprise platform with a typed backend of nearly 1,000 files; worked directly with clients to turn requirements into engineering tasks.

**Hotels.ng** June 2024 – January 2026
*Software/DevOps Engineer (part-time)* *Lagos, Nigeria*

· Built a concurrent Python/Celery application distributing AI workloads over RabbitMQ; wrote CI/CD and monitoring tooling.

## PUBLICATIONS

*Quantum Computing in Artificial Intelligence: A Review of Quantum Machine Learning Algorithms*, Path of Science. doi:10.22178/pos.117-25

*Secure and Resilient Industrial IoT Architectures for Smart Manufacturing: A Comprehensive Review*, JERR (2025). doi:10.9734/jerr/2025/v27i61548

*Clinical Applications of 3D-Printed Medical Implants*, JOPIR. doi:10.69613/22dn3m33

## HONOURS, TEACHING AND COMMUNICATION

PTDF Scholarship (2018) for academic excellence.

**Microsoft Certified Trainer** and former **Microsoft Learn Student Ambassador**: trained over 300 students in software engineering and cloud technologies.

**Director of Research, Mechanical Engineering, FUTO**: ran hands-on training in machine learning, microcontrollers and 3D printing for 50+ students and staff. Co-founder of Hack4FUTO.

Technical writer at vicradon.hashnode.dev.